

International Gas Markets & Economic Evaluation of Gas Projects

Strategic, Financial, and Risk-Informed Decision Making
in the Global Natural Gas Industry



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Natural gas plays a pivotal role in the global energy transition—serving as both a base load fuel and a flexible complement to renewables. In addition, natural gas has gained international strategic appeal and is a source of value creation. This course offers a comprehensive framework for understanding international gas markets and evaluating gas investment opportunities under uncertainty. Participants will explore the full value chain—from upstream extraction and LNG processing to transportation, trading, and market pricing—while mastering economic evaluation techniques tailored to gas projects. The program integrates financial modelling, policy economics, and decision analysis tools such as Monte Carlo simulation, probabilistic reserves, and real options. It prepares professionals to make informed investment decisions in a volatile, carbon-constrained world.

Software Requirements: The course uses **Lumivero’s** Decision Suite tools (Risk and PrecisionTree), though alternative modeling approaches will be discussed. No prior software access is required.

 LOCATION

LONDON

BALI

DUBAI

RIO DE JANEIRO



Course Objectives

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By the end of this course, participants will be able to:

- Analyse the structure and dynamics of regional and global natural gas markets, including LNG.
- Apply economic evaluation techniques to upstream, midstream, and LNG projects.
- Quantify uncertainty in reserves, prices, and costs using probabilistic models.
- Use decision trees, Monte Carlo simulation, and real options to evaluate gas investments.
- Assess fiscal regimes, policy incentives, and ESG impacts on gas project economics.
- Develop decision support models that integrate technical, financial, and strategic dimensions.

Who Should Attend

- Gas project analysts and investment managers
- LNG commercial and strategy professionals
- Energy economists and financial modellers
- Policy advisors and regulatory specialists
- Portfolio managers and business developers
- Engineers and planners transitioning into commercial roles



Training Methodology

This course provides comprehensive learning experience that integrates the instructor's firsthand experience in international energy companies with a robust theoretical framework. Drawing from years of practical experience in the oil & gas sector, the instructor brings real-world insights and industry best practices to the discussions.

- Hands-on modelling using Excel and simulation tools
- Case studies from global gas projects (LNG, pipeline, upstream)
- Expert-led discussions on market trends and policy impacts
- Scenario planning and strategic valuation exercises
- Capstone project integrating course methodologies

Day 1

Global Gas Industry Structure & Decision Context

- Natural gas value chain: upstream, midstream, LNG, and downstream
- Regional market structures: North America, Europe, Asia-Pacific
- Global LNG trade flows and pricing mechanisms
- Supply-demand dynamics and geopolitical influences
- Value chain economics and decision framing
- Expected value and decision analysis foundations

Day 2

Financial Modelling & Cash Flow Economics

- Investment decision techniques: NPV, IRR, Payback
- Sensitivity analysis in gas project economics
- Commodity price forecasting: gas hubs, LNG indices, and contracts
- Cash flow modelling for upstream and LNG assets
- Downstream dependence: electricity markets and petrochemicals
- **Case study:** Evaluating economic feasibility of a floating LNG project

Day 3

Uncertainty Analysis & Quantitative Methods

- Uncertainty vs. risk in gas project evaluation
- Probabilistic reserves and volumetric modelling
- Risk classifications: market, regulatory, technical, operational
- Monte Carlo simulation for price and cost uncertainty
- Decision trees and real options in gas infrastructure
- **Case study:** Decision tree modelling of hydrocarbon exploration projects



Day 4

Strategic Planning & Policy Economics

- Policy incentives and carbon pricing impacts
- ESG considerations and methane emissions economics
- Fiscal regimes for gas: royalties, taxes, PSCs, and hybrid models
- Strategic responses to market volatility and policy shifts
- Scenario planning and stress testing
- **Case study:** Strategic evaluation of a cross-border LNG terminal

Day 5

Practical Applications & Capstone Presentations

- Integrating economic, financial, and risk frameworks
- Using @Risk and PrecisionTree for gas project modeling
- Evaluating long-term profitability and capital allocation
- Future trends: hydrogen blending, carbon allowance markets, digital trading
- **Capstone project:** Presenting an investment recommendation for a gas asset under uncertainty

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BABAK JAFARIZADEH

Dr. Babak Jafarizadeh is a senior consultant and engineering economist who views engineering as the pursuit of economic solutions to technical challenges in an uncertain world. With extensive experience in the oil and gas industry, he has served as a senior economic analyst supporting multi-million-dollar investment decisions for projects across the globe.

Since 2019, Dr. Jafarizadeh has worked as a senior consultant for leading training and consulting companies, while also holding academic appointments at several prestigious institutions — including Heriot-Watt University in the UK (Assistant Professor), the University of California, Berkeley (Lecturer at the Fung Institute for Engineering Leadership), and previously the University of Greenwich (Senior Lecturer). His earlier industry experience includes serving as a Senior Economic Analyst with Equinor (formerly Statoil) in Norway and the USA.

Dr. Jafarizadeh holds a PhD in Petroleum Engineering with a concentration in Investment and Decision Analysis from the University of Stavanger (in collaboration with the University of Texas at Austin), as well as an MSc in Financial Engineering and a BSc in Industrial Engineering. He is an active member of several professional associations, including the Society of Petroleum Engineers (SPE), the Institute for Operations Research and the Management Sciences (INFORMS), and the European Association of Geoscientists and Engineers (EAGE).

His teaching and research interests lie at the intersection of decision analysis and corporate finance, with particular applications in engineering and the energy industry. He has designed and delivered numerous professional training courses, such as Advanced Financial Modelling for Oil and Gas Professionals, Advanced Project Economics, Oil and Gas: Markets, Trading, Pricing and Economic Frameworks, and Energy Economics. Dr. Jafarizadeh is also the author of the book Economic Decision Analysis (for Project Feasibility Studies) and continues to contribute to both academia and industry through research, consulting, and executive education. His areas of expertise include petroleum economics, investment and decision analysis, project feasibility studies, corporate finance, and financial modelling for energy projects.

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